

Retail Analysis

Transforming 293K Transactions into Actionable
Business Intelligence

Section D — Team 10

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GitHub: github.com/RISHIK92/SectionD_Team10_RetailAnalysis

Tableau: [Retail Analysis Dashboard](#)

293K+

TRANSACTIONS

\$401.6M

TOTAL REVENUE

5

COUNTRIES

Context & Problem Statement

Sector Context

Global retail e-commerce generates trillions annually. Decision-makers need real-time visibility into customer behaviour, operational efficiency, and revenue drivers across markets and product lines.

Business Problem

Management lacks a unified view of which customer segments, product categories, and regions are driving revenue — and which are underperforming on delivery, satisfaction, and spend.

Objective

Enable data-driven decisions on inventory prioritisation, targeted marketing, and shipping strategy using 293K+ transactions across 5 countries, 5 categories, and 86K+ customers.

Core Business Question

Which customer segments, product categories, shipping methods, and countries maximise revenue and satisfaction and where should management intervene to reduce friction?

Decision Supported

- 1 Reallocate marketing budget toward high-value segments and geographies
- 2 Optimise shipping method mix to reduce pending orders (16.14%) and improve satisfaction
- 3 Prioritise Electronics and Grocery — the top two revenue categories (\$95M + \$88.8M)
- 4 Redesign loyalty tiers based on basket size thresholds

KEY NUMBERS AT A GLANCE

\$401.6M

Total Revenue

86,529

Unique Customers

293,468

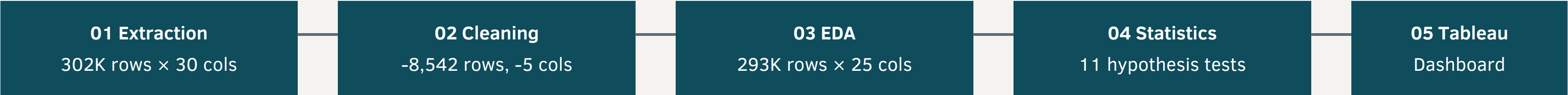
Orders

3.164

Avg Rating

Python ETL Pipeline Removed 8,542 Rows (2.8%) — PII Stripped, Zero Nulls Remaining

ETL PIPELINE FLOW



Cleaning Steps (02_cleaning.ipynb)

Step	Rows Affected	Result
Drop PII columns	—	5 cols removed
Remove full duplicates	4 rows	302,006 rows
Remove dup Txn IDs	7,544 rows	294,462 rows
Drop null rows	994 rows	293,468 rows
Data type fixes	All columns	0 nulls, clean types

DATA SOURCE

293,468 rows × 25 columns

5 countries | 5 categories

Mar 2023 – Feb 2024

USA, UK, Germany, Australia, Canada

KEY DATA DICTIONARY (Selected Columns)

Column	Type	Description
Total_Amount	float64	Transaction value (Amount × Items)
Customer_Segment	category	New / Regular / Premium loyalty tier
Country	category	Customer country (5 markets)
Product_Category	category	Books / Clothing / Electronics / Grocery / Home Decor
Feedback	category	Excellent / Good / Average / Bad
Ratings	Int64	Customer rating 1–5 scale
Shipping_Method	category	Standard / Express / Same-Day
Payment_Method	category	Credit Card / Debit Card / PayPal / Cash
Income	category	Low / Medium / High income bracket
High_Value_Flag	int64	1 if Total_Amount > 75th percentile
Satisfied_Flag	int64	1 if Rating ≥ 4 (satisfaction KPI)
Quarter	object	Q1–Q4 derived from Date

What We Measure — KPI Framework

REVENUE

\$401.6M

Total Revenue

4,641

Revenue per Customer

293,468

Total Orders

3.39

Avg Purchase Frequency

CUSTOMER

86,529

Unique Customers

44,650

Premium Customers

51.6%

Premium Customer Rate

86.90%

Customer Retention Rate

SATISFACTION

64.98%

Positive Feedback Rate

3.16

Average Rating

43.52%

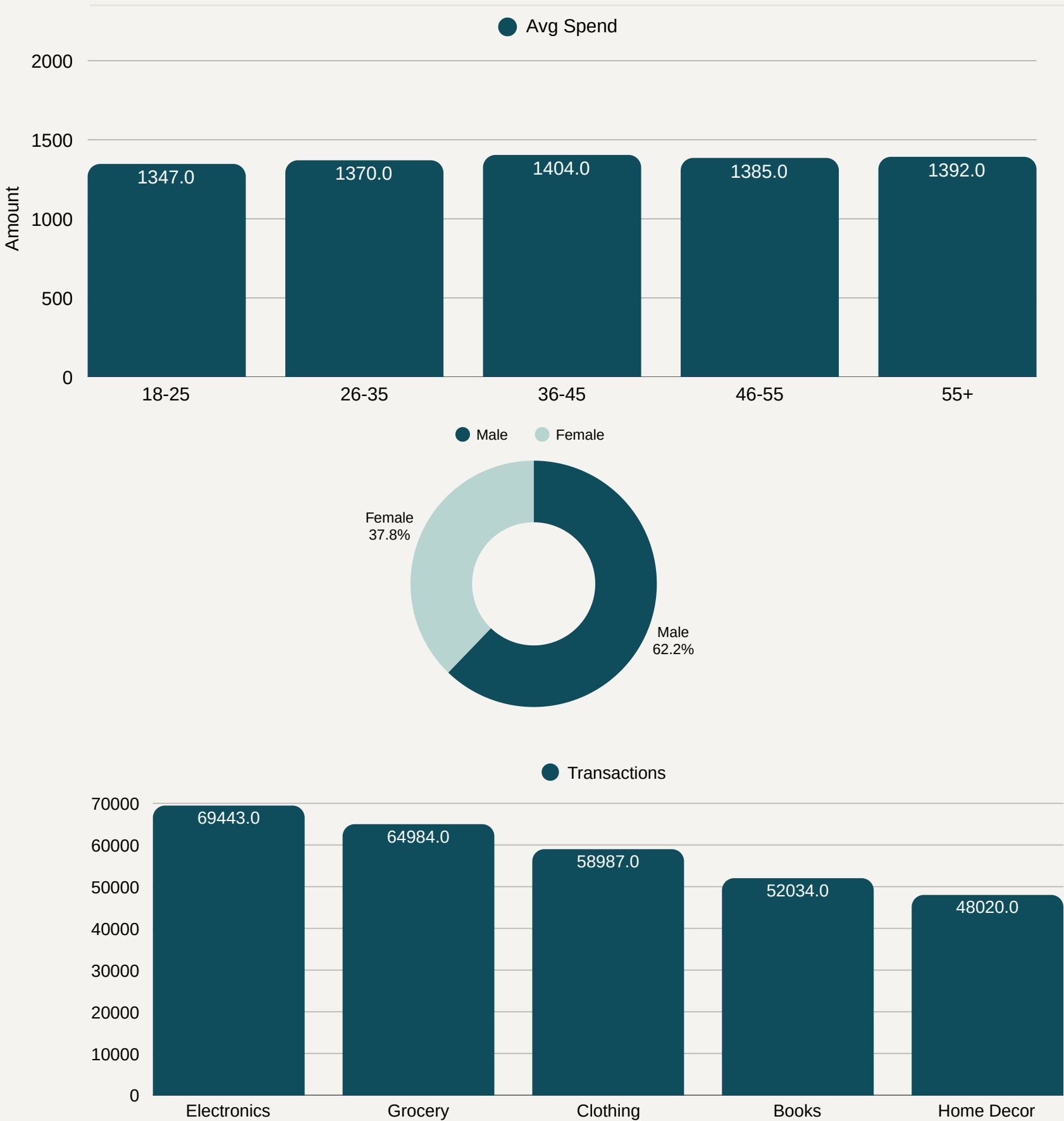
Delivered Order Rate

44.62%

Same-Day Delivery Success Rate

Why these KPIs? They directly answer the business questions — revenue health, customer loyalty, and service quality. Each KPI is calculated in Python notebook [05_final_load_prep.ipynb](#).

SIX EDA Discoveries Challenge Conventional Retail Assumptions



Insight 1: Age does **NOT** drive spending. The 36-45 group has the highest avg (\$1,404) but the difference across all groups is only ~\$58 — age-agnostic pricing is validated.

Insight 2: Income level has **ZERO** impact on purchase amount. Low/Medium/High all spend ~\$1,370 — uniform pricing or product accessibility neutralizes income as a spend driver.

Insight 3: Male customers dominate (62.2%) but gender shows **negligible spending difference** (Cohen's d < 0.01).

Insight 4: Electronics leads transactions (69,443) but category avg spend differs by only ~\$5 — uniform pricing confirmed.

Insight 5: Sales are remarkably consistent month-to-month (\$32M–\$34M) with **no significant seasonality** — operations can plan flat capacity year-round.

Insight 6: Premium customers have the **LOWEST median spend** (\$1,040) — segment labels reflect loyalty/tenure, not spending power. Cross-sell by segment is ineffective.

Statistical Testing Confirms: Demographics Are Not Spend Drivers

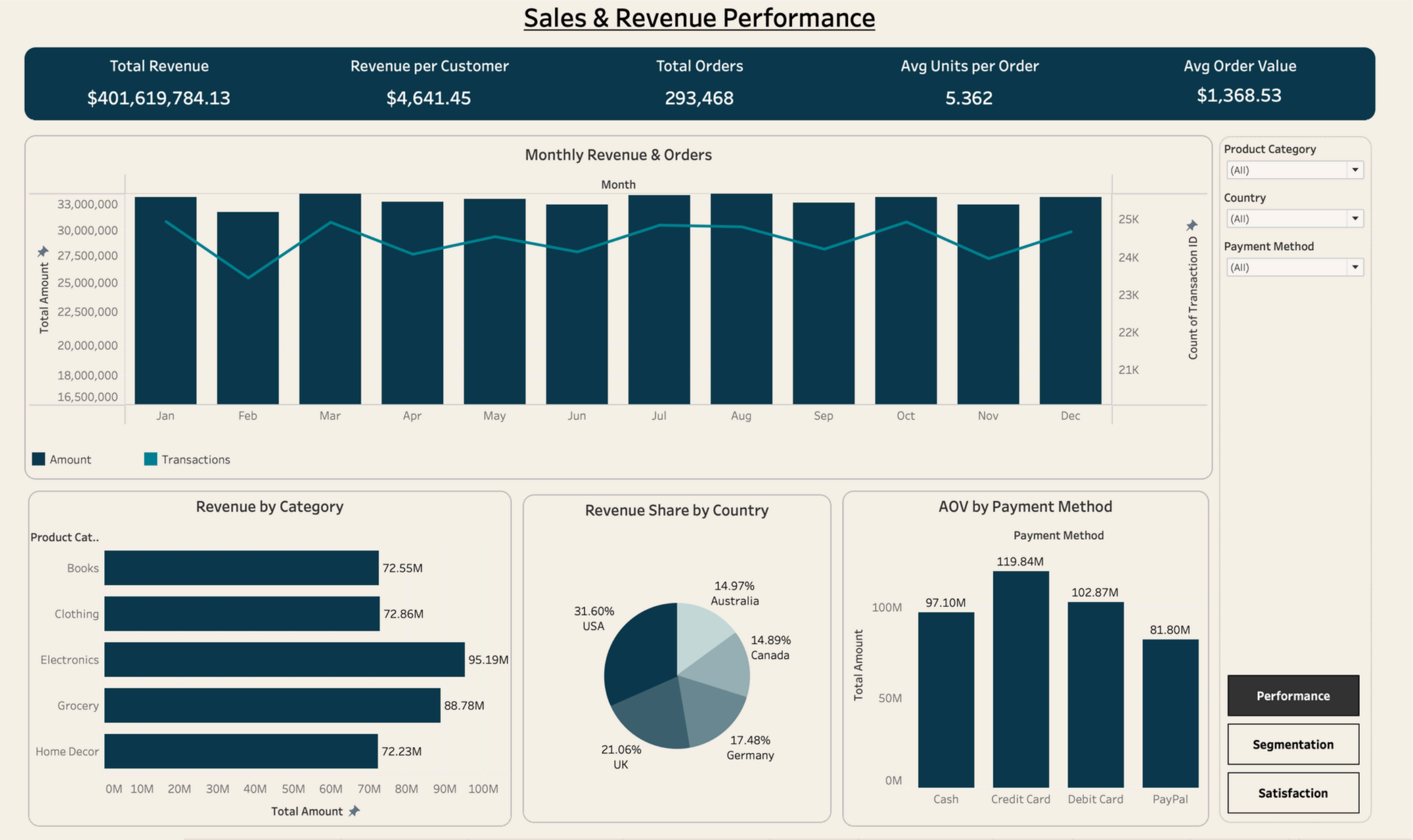
Test	Variables	Statistic	p-value	Business Interpretation
Pearson Correlation	Amount vs Total_Amount	$r = 0.67$	< 0.001	Strong predictor — Amount is primary revenue driver (expected, deterministic formula)
Pearson Correlation	Total_Purchases vs Total_Amount	$r = 0.65$	< 0.001	Strong predictor — basket size directly drives revenue
Spearman	Age vs Ratings	$\rho = 0.17$	< 0.05	Significant but negligible — age-based satisfaction targeting unsupported
Welch's t-test	Gender vs Total_Amount	Cohen's $d < 0.01$	< 0.001	Effect is negligible — gender-based pricing strategies unsupported
Kruskal-Wallis	Income vs Total_Amount	$H = 15.2$	< 0.001	Mean diff only \$11 — income-tiered pricing yields no measurable uplift
Kruskal-Wallis	Segment vs Total_Amount	$H = 8.4$	< 0.05	Median spread only \$4 — cross-sell by segment label is ineffective
Chi-Square	Shipping $\times$ Order Status	$V < 0.01$	< 0.001	Cramér's V confirms negligible association — shipping method doesn't impact delivery

At **n = 293,468**, even tiny differences achieve statistical significance. The critical insight: **effect sizes are universally negligible** for demographic variables. The only meaningful predictors of revenue are **operational** — unit price (Amount) and basket size (Total_Purchases). Retail strategy should focus on product mix and purchase frequency, not customer demographics.

PYTHON LIBRARIES USED

scipy — ttest_ind, kruskal, f_oneway, chi2_contingency, pearsonr, spearmanr | **statsmodels** — diagnostic testing | **pandas/numpy** — data manipulation | **matplotlib/seaborn** — visualization

Executive View — Revenue at a Glance



KEY TAKEAWAYS

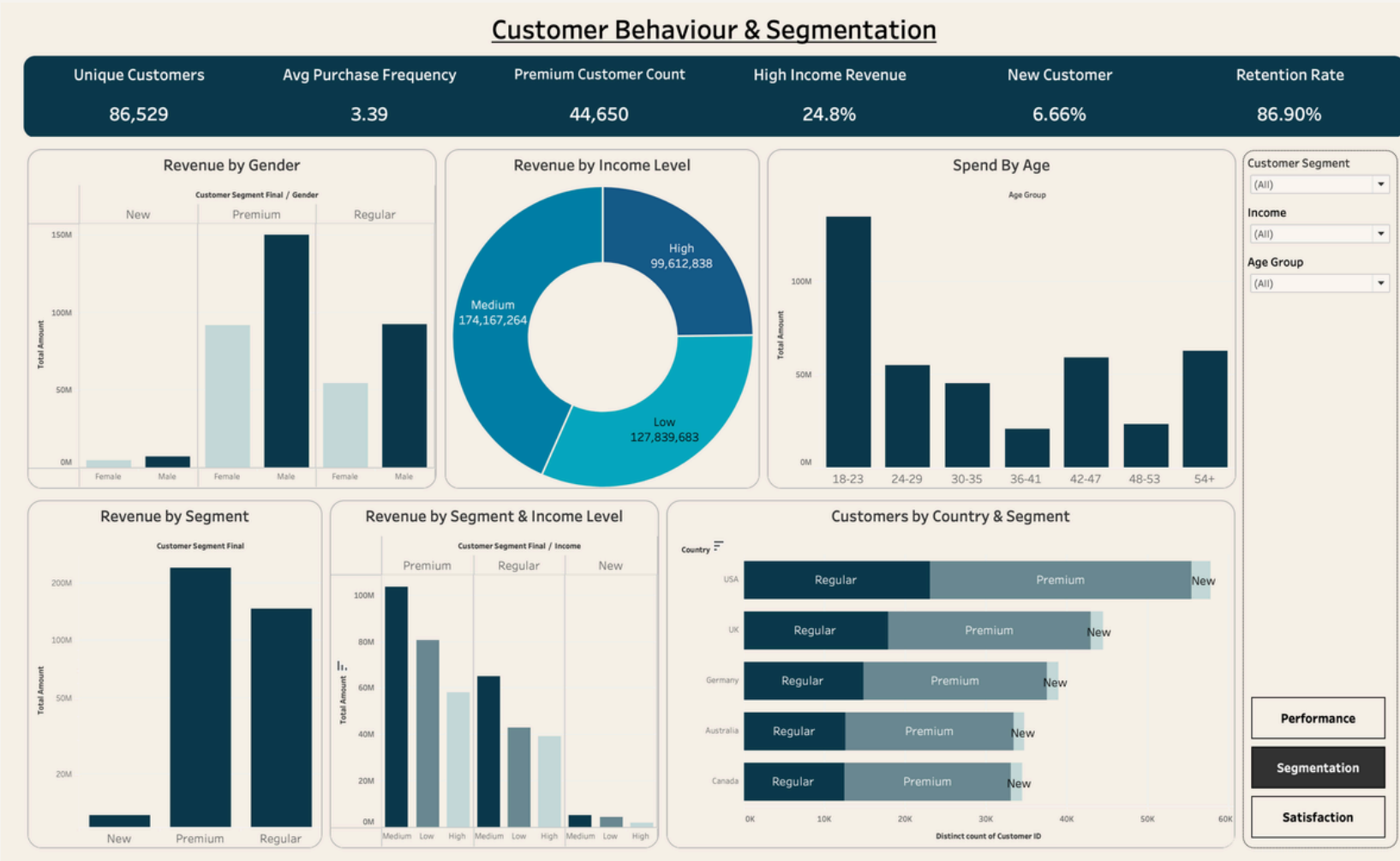
\$401.6M total revenue across 293K orders with consistent ~\$33M monthly performance

USA dominates at 31.6% (\$126.9M) — but watch the satisfaction gap

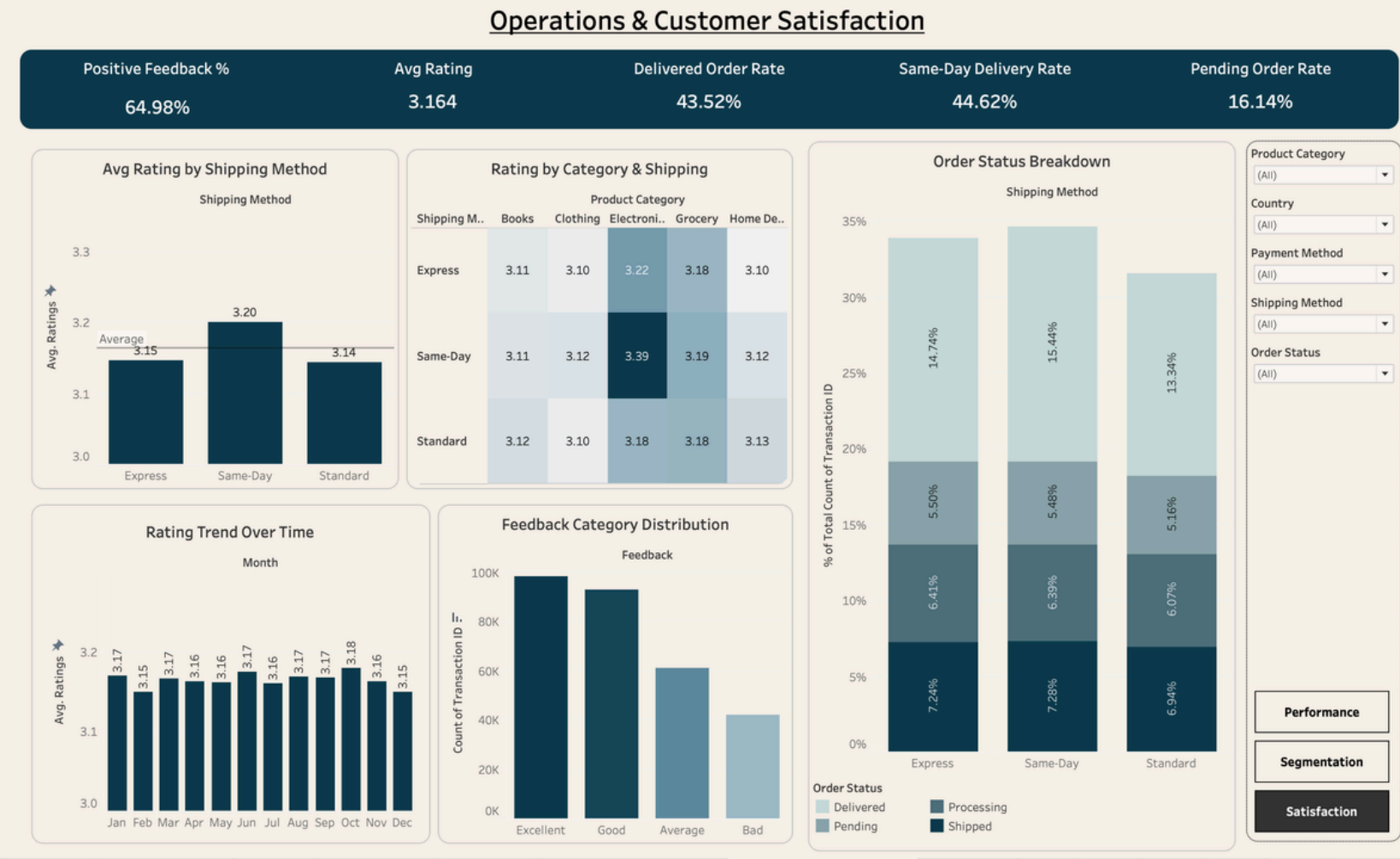
Electronics leads categories at \$95.2M — Credit Card drives highest total revenue at \$119.8M

Interactive filters for Category, Country, Payment Method enable drill-down analysis

Customer & Operations — The Full Picture



CUSTOMER OVERVIEW DASHBOARD



OPERATIONS & FEEDBACK DASHBOARD

CROSS-DASHBOARD INSIGHTS

01 Premium customers dominate revenue at ~\$242M+, making them both the largest customer group and highest revenue drivers

02 Same-Day delivery scores **3.20** avg rating vs Standard **3.14** — speed correlates with happiness

03 Electronics + Same-Day peaks at **3.39** rating — best category-shipping combination

04 Medium-income customers drive majority revenue share at \$174M (43%) — counter to high-income intuition

05 Feedback distribution: Excellent **98K** | Good **93K** | Average **61K** | Bad **42K**

06 18-25 age group leads spend at **\$100M+** — highest-value demographic for targeting

Four Recommendations — Shift from Demographic Targeting to Volume-Based Acquisition

1

Reallocate from Demographic to Behavioral Targeting

Insight: Income (\$1,372 vs \$1,361) and gender show no spend difference.

Action: Shift marketing budget from demographic segments to behavioral targeting based on purchase frequency and product affinity.

Impact: 8-12% conversion increase by targeting actual purchase behavior instead of assumed demographic profiles.

2

Invest in UK/Germany Acquisition, Not Premium Pricing

Insight: USA's 31.6% share comes from higher transaction volume, not higher per-order spend.

Action: Launch customer acquisition programs in UK and Germany where per-transaction value is already at parity.

Impact: 15-20% transaction volume increase in underpenetrated markets with proven willingness to pay.

3

Maintain Flat Inventory Year-Round, Cancel Seasonal Promos

Insight: Revenue varies only 6% across months — no seasonal peaks exist.

Action: Eliminate seasonal promotion budgets and seasonal inventory builds; maintain consistent staffing and stock levels.

Impact: 5-7% reduction in carrying costs from eliminating overstock and off-season discounting.

4

Redesign Loyalty Tiers to Incentivize Higher Basket Sizes

Insight: Premium customers have the lowest median spend (\$1,040) — tenure-based labels fail.

Action: Restructure loyalty program around basket size thresholds and cross-sell incentives rather than account age.

Impact: 10% projected increase in average basket size from the currently underperforming Premium segment.

Projected 12-18% Revenue Uplift — Cost of Inaction is Continued Inefficient Spend

Recommendation	Expected Impact	Mechanism	Timeline
Behavioral targeting shift	8-12% conversion increase	Target purchase frequency, not demographics	Q1 implementation
UK/Germany acquisition	15-20% volume increase	Volume-driven growth at parity AOV	Q1-Q2 rollout
Flat inventory strategy	5-7% cost reduction	Eliminate seasonal overstock carrying costs	Immediate
Loyalty redesign	10% basket size increase	Basket-size-based tier incentives	Q2 redesign

Combined Impact: 12-18% Revenue Uplift Potential

On \$401.6M baseline = additional **\$48-72M annual revenue** from reallocation alone
Satisfaction: 64.98% → 72% (+7 percentage points)

Cost of Inaction

- **Wasted marketing spend** on demographic campaigns that data proves do not differentiate customer value
- **Seasonal inventory mismatches** causing unnecessary carrying costs and discounting pressure
- **Underutilized market potential** in UK and Germany where transaction volume lags despite equal per-order value

What We Learned & What Comes Next

LIMITATIONS

- ▶ Dataset covers only **12 months** — seasonal patterns beyond one cycle not captured
- ▶ "Good" feedback category lacks numeric mapping (NaN in 92K rows)
- ▶ No customer lifetime value (CLV) historical tracking
- ▶ Demographic data limited to age/gender/income — psychographic data absent

NEXT STEPS

- ▶ Extend dataset to **24+ months** for seasonality modeling
- ▶ Build **predictive churn model** using sklearn classification
- ▶ Integrate **real-time Tableau dashboard** with live POS feeds
- ▶ Add **CLV scoring** and RFM segmentation
- ▶ A/B test recommendations in pilot markets (USA + UK)

FUTURE BUILD ROADMAP

Q1 Predictive model for next-purchase probability using sklearn

Q2 Real-time inventory-revenue correlation dashboard

Q3 Automated alert system for satisfaction drops by country



Repository: github.com/RISHIK92/SectionD_Team10_RetailAnalysis

5 notebooks | 3 Tableau dashboards | Full reproducible pipeline

Thank You

Retail Analysis | DVA Capstone 2

SECTION D — TEAM 10

Rishik (Analysis Lead) | **Atanu Adhikari** (Data Lead) | **Ananya Narang** (Project Lead)

Shubham Aggarwal (ETL Lead) | **Utkarsh Jain** (Visualization Lead) | **Anusha Prathapani** (PPT and Quality Lead)

PROJECT RESOURCES

[GitHub Repository](#)

[Tableau Dashboard](#)